

Requests Responses and JSON

Project: Verified Study Buddy

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Tools and APIs

Without opening the old lesson, explain one key idea from Tools and APIs. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

User Need and AI Contribution

Without opening the old lesson, explain one key idea from User Need and AI Contribution. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Verify Numbers and Dates

Without opening the old lesson, explain one key idea from Verify Numbers and Dates. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Separate failures for JSON and interface contracts. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for JSON and interface contracts. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Specify independently

Define a JSON contract for {"label": "Box", "count": 4}: nonempty text label, integer count 0 through 10. Give a valid case, a wrong-type case and a missing-field case with expected acceptance.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Request id=7 returns status=200 and {"id":8,"pages":90}. Explain why a successful status is insufficient and specify the handling.

6. Solve the given case

Request is for B7; response is {"id":"B7","pages":120}. Another response has malformed JSON.

What checks precede use?

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

An API returns `{"ok":false,"error":"not_found"}`. Your app tries to display `response.title`. What should it do instead?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 22 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Use a stable request/response contract between interface, retriever, model, and verifier.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Request id=7 returns status=200 and {"id":8,"pages":90}. Explain why a successful status is insufficient and specify the handling.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Translate a value

What Python value does JSON null become? What JSON value represents Python True?

My reasoning and evidence

13. W2 Separate failures

Classify {"minutes": "5"} and {minutes: 5} as valid JSON text or malformed JSON.

My reasoning and evidence

14. W3 Repair the interface

A parser accepts a JSON list then crashes on .get. What check belongs before field access?

My reasoning and evidence

15. W4 Specify independently

Define a JSON contract for {"label": "Box", "count": 4}: nonempty text label, integer count 0 through 10. Give a valid case, a wrong-type case and a missing-field case with expected acceptance.

My reasoning and evidence

16. W5 Limit parsing

A JSON record parses successfully. Does that prove its observation is correct?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Requests Responses and JSON

A lunch-count response must satisfy a declared JSON contract.

The toy response has exactly class_code text and portions integer 0..30. Example:

```
{"class_code": "R8", "portions": 12}.
```

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Requests Responses and JSON

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read fields

class_code is nonempty text; portions is the integer 12.



2. Check range

$0 \leq 12 \leq 30$, so the declared count is within range.



3. Accept structure

Accept this response shape; acceptance does not prove that 12 portions were actually counted.

CHANGE ONE THING

Replace 12 with the text "12".

Explain the changed result using the matching lesson.

Would converting the text silently prove the source count is correct? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Verified Study Buddy

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A lunch-count response must satisfy a declared JSON contract.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Accept this response shape; acceptance does not prove that 12 portions were actually counted.

1. Accept structure
2. Read fields
3. Check range

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Accept structure Accept this response shape; acceptance does not prove that 12 portions were actually counted. Read the picture in order. Accept this response shape; acceptance does not prove that 12 portions were actually counted. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: Replace 12 with the text "12". Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. Reject the response under this strict contract.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Reject the response under this strict contract. Digits inside quotes represent text, whereas portions requires an integer.

Repair a missing building block

Review Tools and APIs (ai-3/21). Goal: Explain the mechanism and terminology of tools and api contracts. Short practice: Does a valid C7 ID permit the extra lending action? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review User Need and AI Contribution (ai-3/19). Goal: Explain the mechanism and terminology of capstone requirements and corpus. Short practice: What non-AI baseline would reveal whether this helper adds value? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Verify Numbers and Dates (ai-3/16). Goal: Explain the mechanism and terminology of claim support and citation evaluation. Short practice: Which part can be corrected immediately from existing evidence, and which needs follow-up?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
